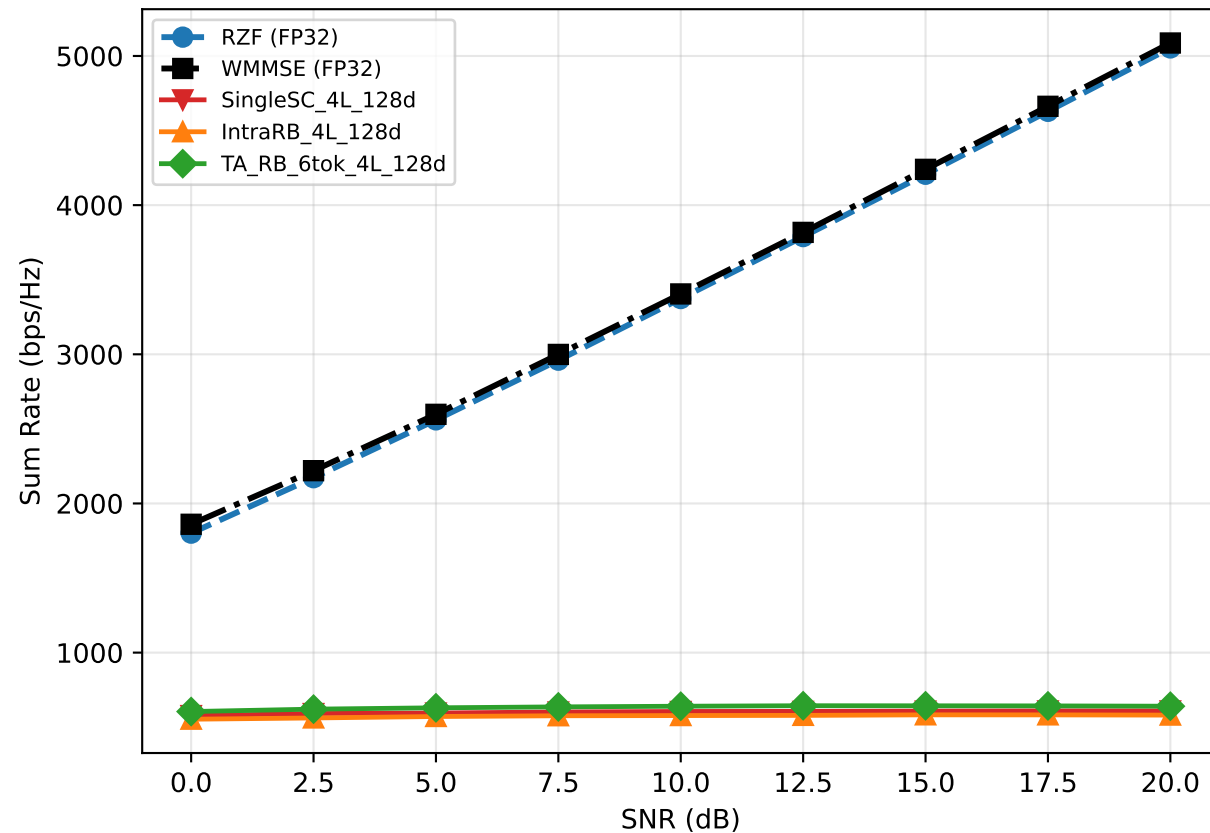
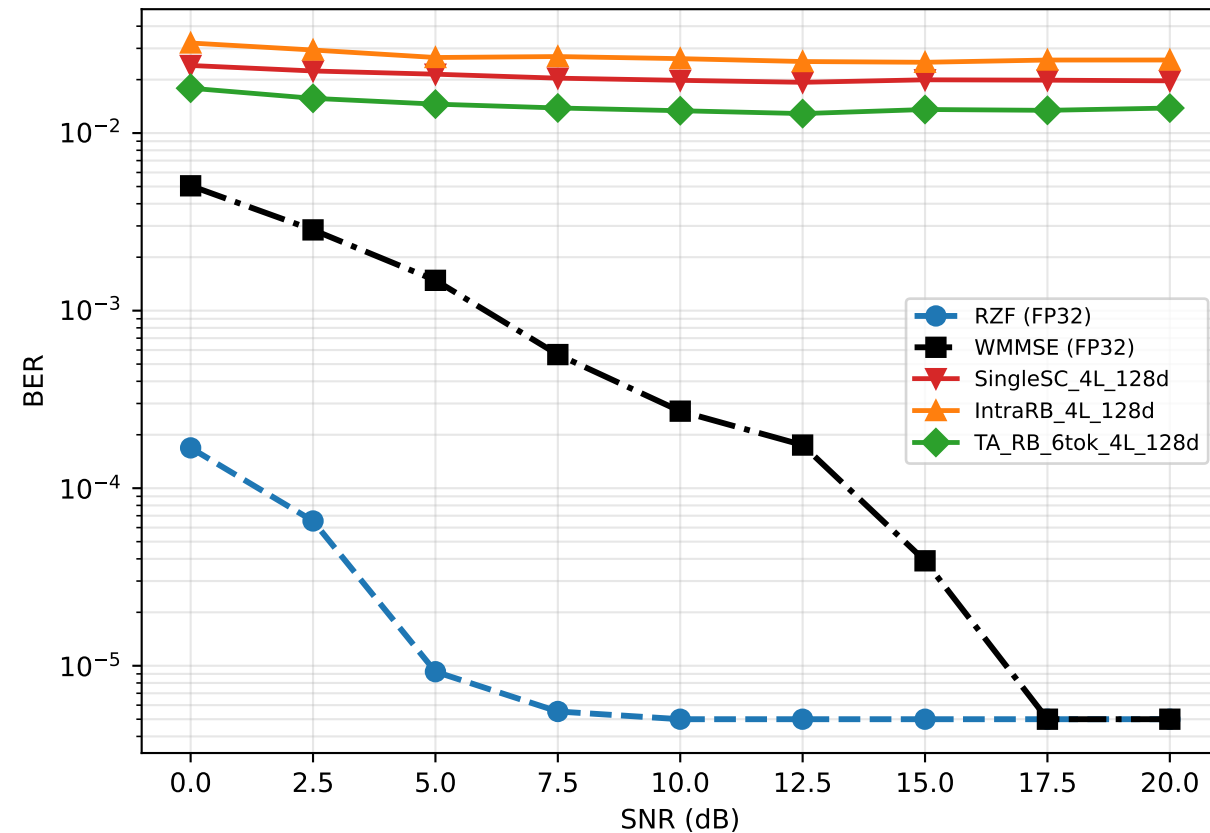


IntraRB Transformer Precoder — 8×4 MU-MIMO (8TX, 4UE, 72SC, SNR train=15dB)

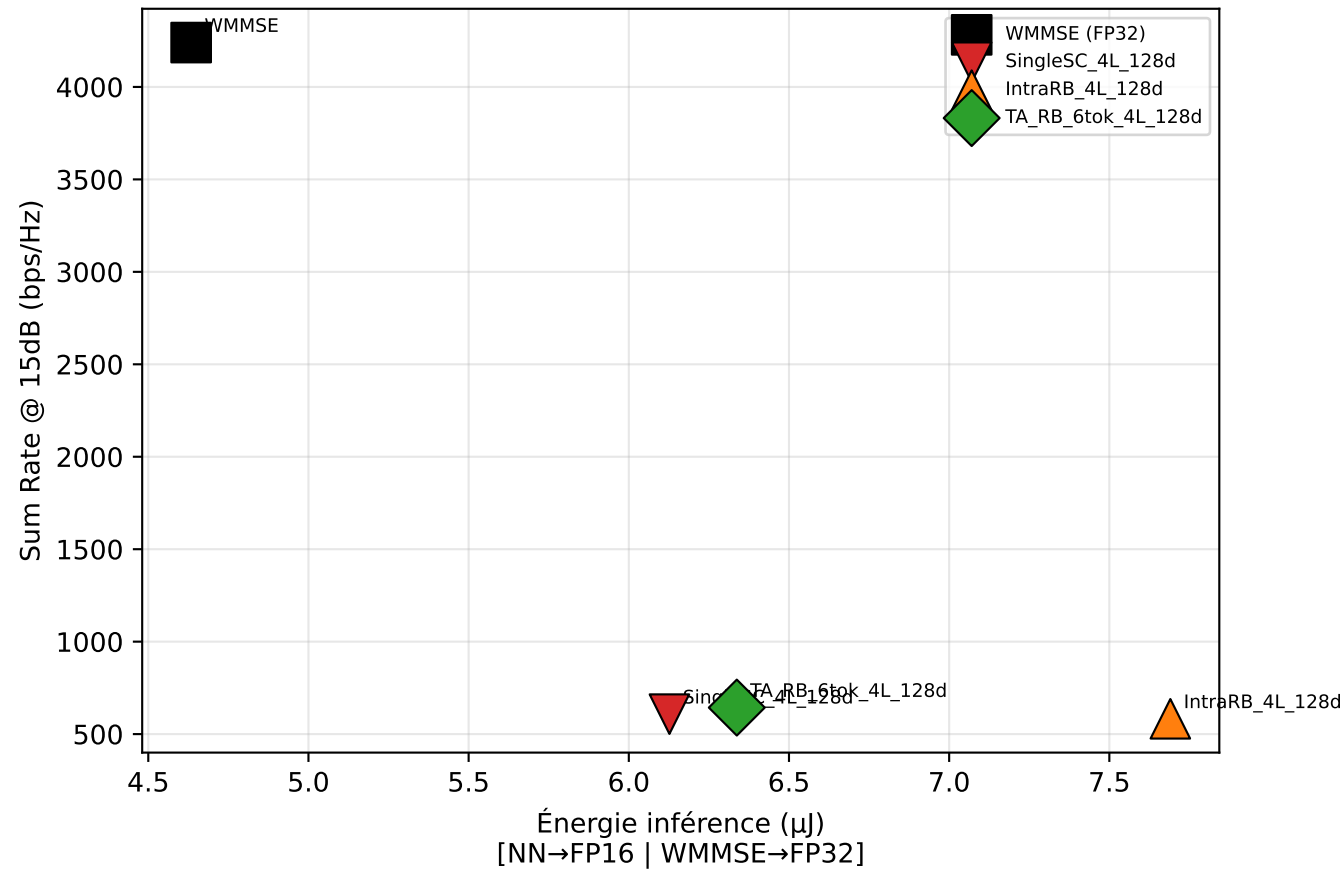
(a) Efficacité spectrale



(b) BER Monte Carlo



(c) Pareto Énergie-Performance



(d) Résumé @ 15dB

Méthode	Rate (bps/Hz)	Gap vs WMMSE	FLOPs (M)	E FP16 (μJ)	Gain FP16
WMMSE (FP32)	4239.97	+0.00	2457	N/A	ref
RZF (FP32)	4206.95	-33.02	3	N/A	ref
TA_RB_6tok_4L_128d	643.42	-3596.54	11735	6.3371	0.7×
SingleSC_4L_128d	607.88	-3632.09	11353	6.1268	0.8×
IntraRB_4L_128d	582.83	-3657.14	14237	7.6905	0.6×